Proof by Induction-Sequences

Key Points

Remember to prove a statement holds for all positive integers by induction, we do the following:

- Show it holds for the base case 1
- Assume it holds for a case k
- Prove the statement then holds for the case $k+1$

Consider a sequence defined by a recurrence relation, for example $a_{n+1} = f(a_n)$ where a_1 is given. To show an inductive step we could start with a_{n+1} and use the recurrence relation to obtain a_n's

Basic Skills

Q1. Given that $a_{n+1} = 3a_n - 1$, and that $a_1 = 1$, calculate

a) a_2 **b)** a_3 **c)** a_4 **d)** a_5

Q2. Given that $a_{n+1} = a_n^2 - a_n - 1$, and that $a_1 = 3$, calculate

a) a_2 **b)** a_3 **c)** a_4

Q3. Suppose that $a_{n+1} = 2a_n - 1$. Given that $a_k = 1$, show that $a_{k+1} = 1$

Q4. Suppose that $a_{n+1} = a_n(a_n - 2)$. Given that $a_k = 4$, show that $a_{k+1} = 8$

Q5. Suppose that $a_{n+1} = 3a_n$. Given that $a_k = 3^k$, show that $a_{k+1} = 3^{k+1}$

Competency Skills

Q6. Let $a_1 = 1$ and $a_{n+1} = a_n + 5$. We want to show the formula $a_n = 5n - 4$ holds for all positive integers n, using a proof by induction

a) Verify that the formula holds for the base case $n = 1$

b) Assume that $a_k = 5k - 4$ for some fixed number k. Show that $a_{k+1} = 5(k+1) - 4$.

c) Write out the statement we can conclude from parts a) and b) using proof by induction

Q7. Let $a_1 = 2$ and $a_{n+1} = 2a_n + 5$. Use a proof by induction to show that $a_n = 7 \times 2^{n-1} - 5$, for all positive integers n.

Q8. Let $a_1 = \frac{3}{2}$ and $a_{n+1} = \frac{a_n^2+2}{3}$. Use a proof by induction to show that $1 < a_n < 2$, for all positive integers n.

Advanced Skills

Q9. Let $a_1 = 1$ and $a_{n+1} = (n+1)a_n$

a) Show that if $a_k > 2^k$ for some positive integer k, then $a_{k+1} > 2^{k+1}$

b) Your friend concludes that $a_n > 2^n$ for all positive integers by induction. Why are they wrong?

Q10. Consider n distinct balls in a row, from which we want to choose r balls, where $0 \leq r \leq n$

a) Give an expression for the number of ways to choose r balls

b) Let $n = k+1$. How many ways are there to choose r balls which must contain a special ball X

c) Let $n = k+1$. How many ways are there to choose r ball which must not contain the special ball X

d) Hence deduce pascal's rule $\binom{k+1}{r} = \binom{k}{r-1} + \binom{k}{r}$

e) Hence prove the binomial coefficient formula $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

Extension

Q11. Let $f(x) = \sin(2ax)$. Show that $\frac{d^n f}{dx^n}(x) = (-4)^{\frac{n}{2}} a^n \sin(2ax)$, for all positive even integers n.

Hint: Can you adjust the method of proof by induction to show what you want?